

# Technical change and health: Evidence from the reduction of sugarcane burning in Brazil

Leonardo Cardoso<sup>1</sup>   Liqing Li<sup>2</sup>   Deyvid Leite<sup>3</sup>   Daiana Wischral<sup>1</sup>   David Just<sup>4</sup>

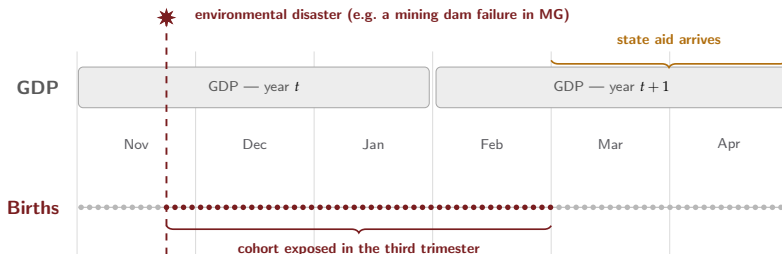
<sup>1</sup>Universidade Federal de Viçosa   <sup>2</sup>Texas A&M University

<sup>3</sup>Universidade Federal de Uberlândia   <sup>4</sup>Cornell University

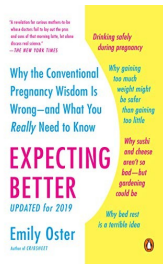
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# Health at birth: a thermometer that reads every day



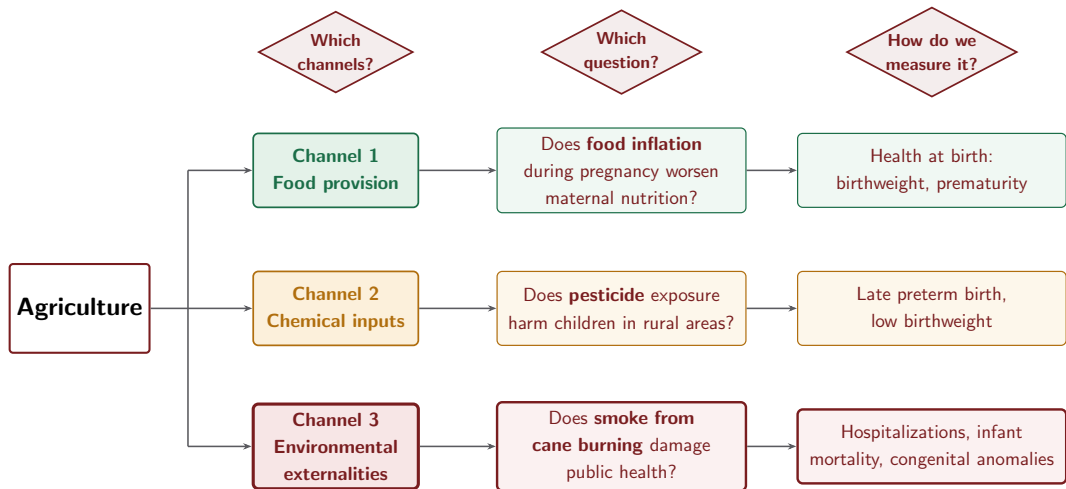
- GDP is a great measure of development, but it arrives once a year
- Babies are born all the time — in 2024, **64** a day in Salvador, **46** in Curitiba, **2** even in Viçosa-MG



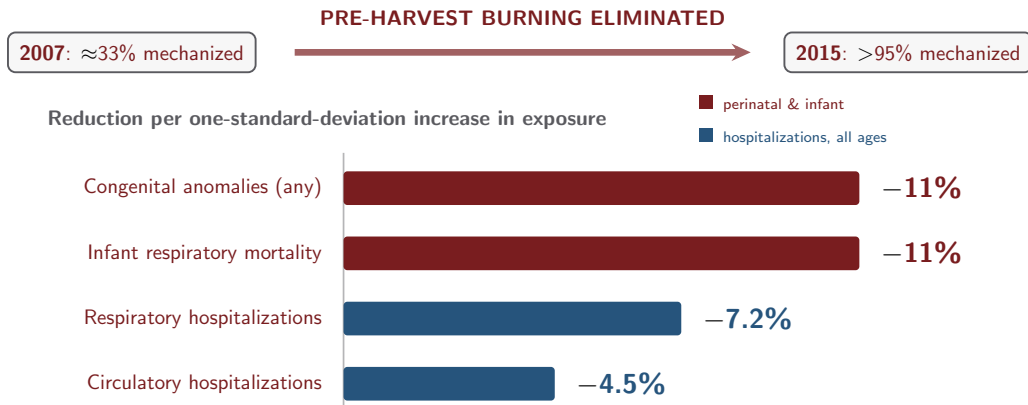
Emily Oster (Brown),  
*Expecting Better* (2013)

Birth records are high-frequency, precisely dated, and cover every birth — a thermometer of local conditions.

# How does agriculture affect health?



# Can a technical change in agriculture improve health?

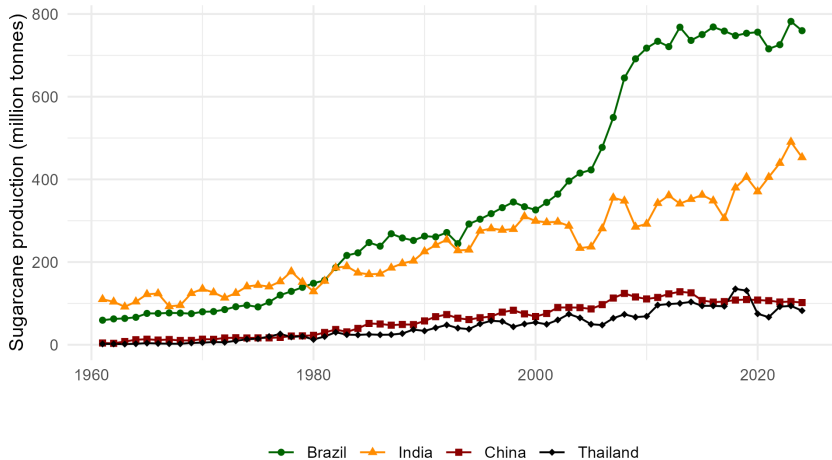


And it really is the smoke:

✓ pollution fell **only** in harvest months

✓ **downwind** municipalities gained most

# Context

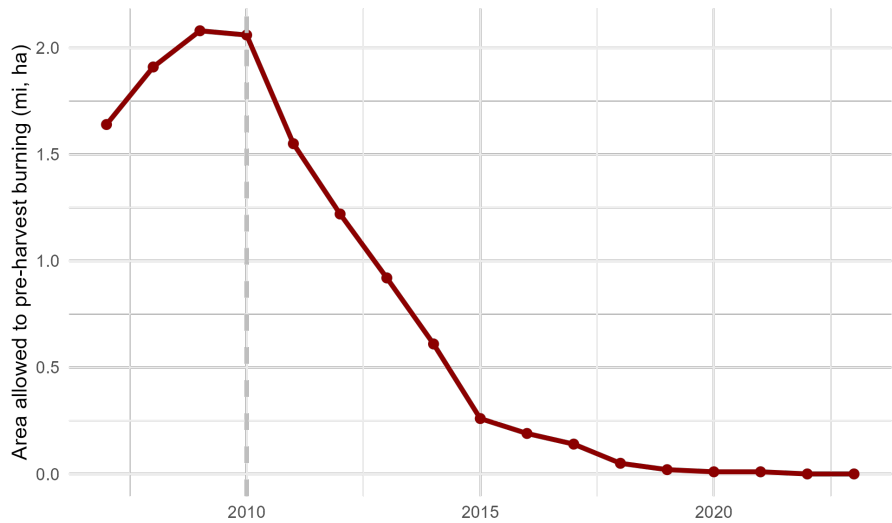


# Our Source of Identification

# There was a reduction in burning driven by...

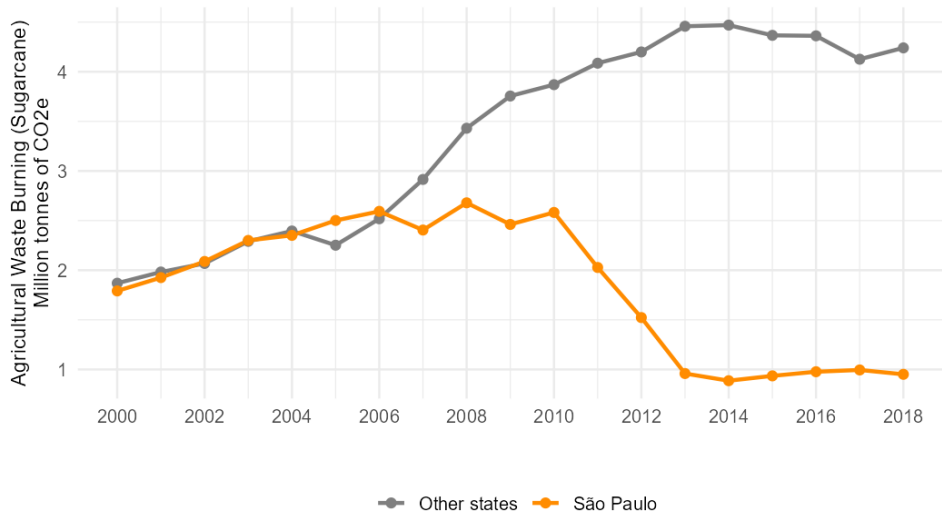
- Law
  - ▶ State Law 11.241/2002: gradual ban on straw burning (targets: 50% by 2011, 100% by 2021)
- Protocol
  - ▶ Green Ethanol Protocol (2007): voluntary industry agreement that accelerated deadlines to 2014–2017 in exchange for preferential financing and export market access
- Labor costs
  - ▶ Rising rural wages and export sustainability standards reinforced the shift [Davis, 2017]

# Area allowed to pre-harvest burning

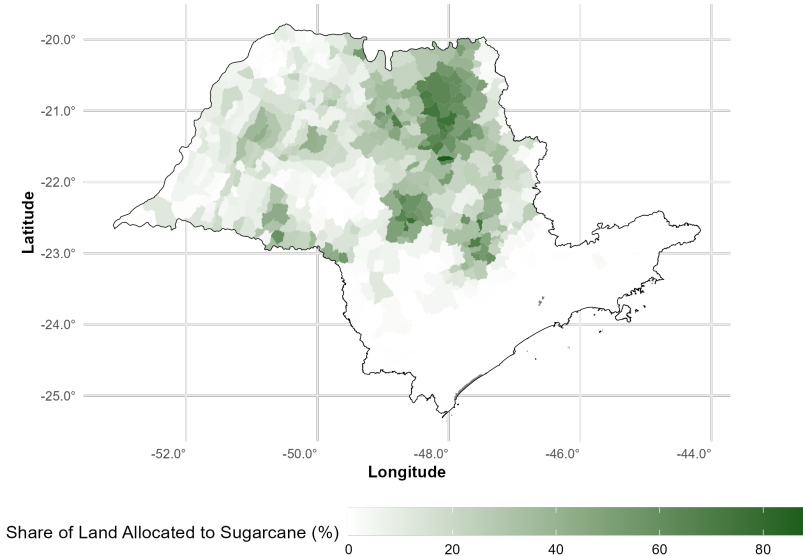




# Emissions from sugarcane agricultural waste burning



# With spatial heterogeneity in this reduction...



# Combining the timing and the spatial heterogeneity

## Difference-in-differences, continuous exposure

$$\ln(Y_{it}) = \alpha \cdot \underbrace{Land_i \times Post_t}_{\text{Tech. Change (TC)}} + \theta_i + \gamma_t + \varepsilon_{it}$$

$TC_{it} = Land_i \times Post_t$ , with  $Post_t = \mathbf{1}[t > 2010]$  and  $Land_i$  = pre-2011 sugarcane land share

- $\theta_i, \gamma_t$ : municipality and year FE; weighted by population (or births); SEs clustered by municipality
- $SD(Land_i) = 0.19$  — the unit for all “one-SD” effects below
- We also run specifications on individual birth records with mother fixed effects

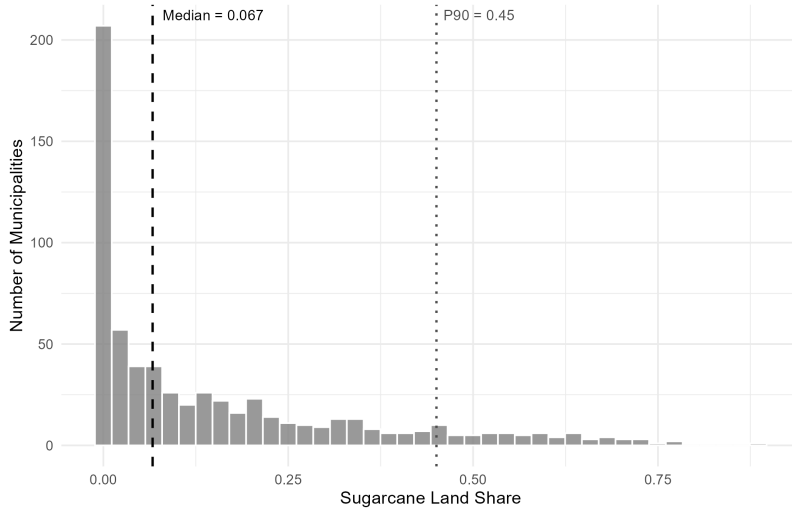
# The Literature

- Rangel and Vogl [2019] – Review of Economics and Statistics: wind-based smoke exposure → lower birthweight, shorter gestation (São Paulo)
- Chagas et al. [2016] – Regional Science and Urban Economics: spatial DiD, sugarcane → respiratory hospitalizations, treatment is a binary median-split

## Our contribution

- Broader outcomes: circulatory disease, infant mortality, a biologically-grounded congenital anomaly index
- Holding sugarcane production constant, what is the impact of reducing burning?
- Continuous rather than binary exposure, plus three sharper designs (season, wind, dose)

# Heterogeneity matters



# Data Sources

- **Hospitalizations:** SIH/SUS, 2008–2022 (ICD J00–J99, I00–I99)
- **Births:** SINASC, 2007–2023,  $N \approx 9.7\text{M}$  individual records
- **Mortality:** SIM, 2002–2023
- **Treatment:** PAM-IBGE sugarcane land share, pre-2011 average
- **PM10/PM25:** CETESB monitoring network, 2003–2024
- **Fire:** MODIS MCD64A1  $\times$  MapBiomas LULC (cane pixels), Google Earth Engine
- **Wind:** NASA POWER/MERRA-2 reanalysis, Jul–Sep 2002–2022

All sources are administrative and cover the universe of events — no survey sampling error.

# Results

# Hospitalizations Fall for Respiratory & Circulatory Disease

| $y = \ln(\text{hosp. per capita})$    | Full (2008–2021)     |                      | Pre-Covid (2008–2019) |                      |
|---------------------------------------|----------------------|----------------------|-----------------------|----------------------|
|                                       | (1)                  | (2)                  | (3)                   | (4)                  |
| <i>Panel A: Respiratory (J00–J99)</i> |                      |                      |                       |                      |
| Tech. Change                          | −0.380***<br>(0.114) | −0.369***<br>(0.112) | −0.287**<br>(0.111)   | −0.286**<br>(0.111)  |
| <i>Panel B: Circulatory (I00–I99)</i> |                      |                      |                       |                      |
| Tech. Change                          | −0.238***<br>(0.078) | −0.224***<br>(0.075) | −0.234***<br>(0.078)  | −0.222***<br>(0.075) |
| log(GDP/cap) control                  |                      | X                    |                       | X                    |
| Municipality & Year FE                | X                    | X                    | X                     | X                    |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Weighted by population, clustered SEs.

- **1 SD more exposure**  $\Rightarrow$  respiratory  $\downarrow$ **7.2%**, circulatory  $\downarrow$ **4.5%**; both stable when controlling for GDP/capita



# Birth Outcomes: No Effect on Weight, Fewer Anomalies

|               | Any Anomaly<br>(Q00–Q99) | Pol3 Index<br>(NTD+Orofac.+Card.) | LBW<br>(<2,500g)       | VLBW<br>(<1,500g)      |
|---------------|--------------------------|-----------------------------------|------------------------|------------------------|
| Tech. Change  | −0.006 66*<br>(0.003 90) | −0.004 27**<br>(0.001 82)         | 0.001 24<br>(0.002 00) | 0.000 10<br>(0.000 84) |
| Baseline mean | 11.45/1000               | 3.59/1000                         | 9.26%                  | 1.47%                  |

Individual (Sinasc),  $N \approx 9.7\text{M}$ , Controls: Mun., year, educ., sex, marital, race...

- **No effect on birthweight** (LBW, VLBW)
- **Congenital anomalies fall 11%**; the **Pol3 pollution-linked index falls 22.6%** at 1 SD of exposure
- Pol3 = neural tube defects + orofacial clefts + cardiac malformations

# Which Anomalies?

| $y = \text{Pr}(\text{anomaly type})$ | Neural Tube<br>(Q00–Q07)    | Orofacial<br>(Q35–Q37)   | Cardiac<br>(Q20–Q26)        | Pol3 Index                  |
|--------------------------------------|-----------------------------|--------------------------|-----------------------------|-----------------------------|
| Tech. Change                         | −0.000 539**<br>(0.000 221) | 0.000 000<br>(0.000 156) | −0.003 846**<br>(0.001 623) | −0.004 265**<br>(0.001 823) |
| Baseline (per 1,000)                 | 1.09                        | 0.73                     | 1.91                        | 3.59                        |
| Relative effect (1 SD)               | −9.4%                       | ≈ 0                      | −38.3%                      | −22.6%                      |

# Mortality: A Dose-Response Gradient

## Adults — no effect

|              | Respiratory       | Circulatory       |
|--------------|-------------------|-------------------|
| Tech. Change | -0.028<br>(0.093) | -0.043<br>(0.071) |

⇒ -0.5% and -0.8% at 1 SD —  
statistically zero

## Infants (<5) — significant

|              | Resp. mortality    |
|--------------|--------------------|
| Tech. Change | -0.593*<br>(0.344) |

⇒ -11% at 1 SD; all-cause infant mortality:  
null

Infants are physiologically more susceptible.

# Is It Really the Smoke?

# Less Fire and Less Particulate Matter

$$\ln(Y_{it}) = \alpha \cdot \underbrace{Land_i \times Post_t}_{\text{technical change - same treatment as before}} + \theta_i + \gamma_t + \varepsilon_{it}$$

technical change – same treatment as before

$Y_{it} \in \{ \text{burned area on cane pixels, PM10, PM25} \}$

|                                                        | Harvest (Jul–Sep)    | Off-season (Jan–Mar) |
|--------------------------------------------------------|----------------------|----------------------|
| <i>Panel A: Sugarcane burning (MODIS × MapBiomass)</i> |                      |                      |
| $\ln(\text{burned area, cane pixels})$                 | −4.964***<br>(0.450) | —                    |
| <i>Panel B: Particulate matter (CETESB)</i>            |                      |                      |
| $\ln(\text{PM10})$                                     | −0.094***<br>(0.010) | −0.005<br>(0.006)    |
| $\ln(\text{PM25})$                                     | −0.091***<br>(0.010) | −0.001<br>(0.006)    |

\*\*\*  $p < 0.01$ . Off-season burned area remains near zero throughout (placebo).

- Harvest-season burned area on cane fields falls  $\approx 66\%$  at mean exposure

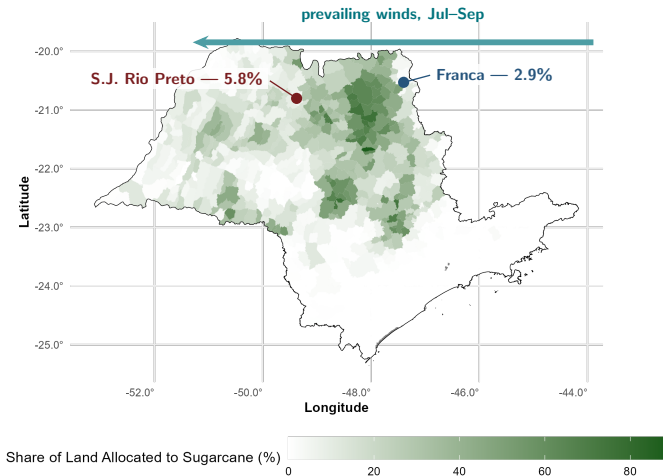
# Health Gains Concentrate in Harvest Months

$$\ln Y_{itm} = \alpha \cdot TC_{it} + \beta \cdot (TC_{it} \times Harvest_m) + \theta_{i \times m} + \gamma_t + \varepsilon_{itm}$$

| $y = \ln(\text{hosp.}/\text{pop})$ | Respiratory          | Circulatory         |
|------------------------------------|----------------------|---------------------|
| Tech. Change (off-season)          | -0.135***<br>(0.033) | -0.083**<br>(0.028) |
| Tech. Change $\times$ Harvest      | -0.069***<br>(0.011) | -0.014<br>(0.013)   |

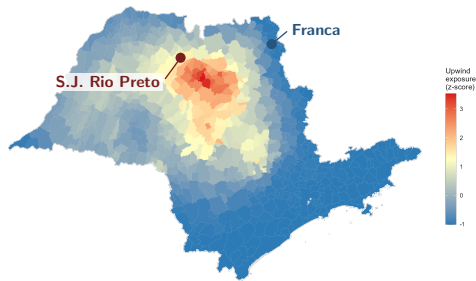
Respiratory hospitalizations fall more when burning would have occurred. Circulatory hospitalizations show no such difference.

# Why the wind matters



$$WindExp_{it} = \sum_j \frac{\max(0, \cos(\hat{w}_{it} - \angle_{i \rightarrow j}))}{dist_{ij}} \cdot SC_j$$

# Smoke Travels with the Wind



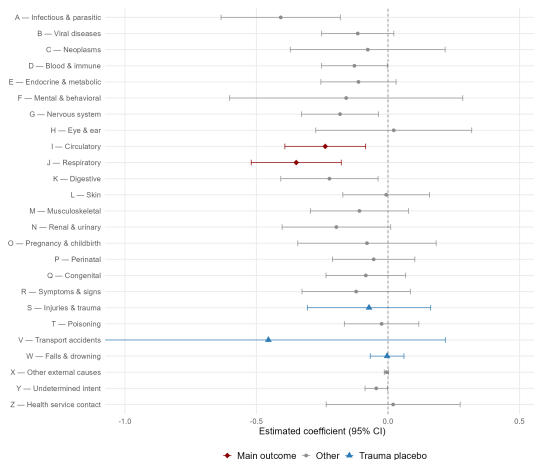
|              | Respiratory |           | Circulatory |           |
|--------------|-------------|-----------|-------------|-----------|
|              | Full        | Pre-Cov.  | Full        | Pre-Cov.  |
| Tech. Change | -0.348***   | -0.265*** | -0.238***   | -0.222*** |
| WindExp      | -0.035*     | -0.027    | -0.033      | -0.011    |

Municipalities downwind of cane fields — e.g. São José do Rio Preto — gain more than upwind ones such as Franca with identical own-exposure. Wind direction is plausibly exogenous to local health trends and, just as importantly, it is what the environmental economics literature now expects.



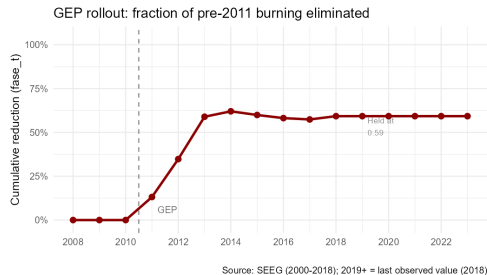
# More Robustness

# Placebo Across All ICD-10 Chapters



If the design were picking up general improvements in municipal health infrastructure, every chapter would move. It does not: only the two chapters with a biological link to particulate matter respond.

# Continuous Treatment Intensity



|                                | Respiratory | Circulatory |
|--------------------------------|-------------|-------------|
| Binary (0/1)                   | −0.344***   | −0.239***   |
| Continuous ( $\times fase_t$ ) | −0.618***   | −0.393**    |

Replacing the post-2010 dummy with  $fase_t$  — the fraction of burning emissions already eliminated — gives an implied effect at the mean of  $-0.346$ , essentially the binary estimate.

## Robustness: Excluding São Paulo City

|               | Respiratory |           | Circulatory |           |
|---------------|-------------|-----------|-------------|-----------|
|               | Full        | Pre-Covid | Full        | Pre-Covid |
| Baseline      | -0.380***   | -0.287**  | -0.238***   | -0.234*** |
| Excl. SP city | -0.327**    | -0.236*   | -0.228**    | -0.189**  |

São Paulo city has 12M inhabitants and zero cane share — a heavily weighted, mechanically untreated observation. Dropping it leaves the estimates essentially unchanged.

# What Are the Stakes?

- Aggregating the hospitalization reduction over 2011–2021 across exposed municipalities: **≈118,000** fewer hospital admissions (71,000 respiratory + 47,000 circulatory)
- At 2019 SUS reimbursement rates: **R\$ 209 million** (USD 53M) in avoided hospitalization costs
- **≈1,640** fewer births with cardiac malformations (2011–2023)  $\Rightarrow$  **R\$ 22 million** in avoided surgical costs

**R\$ 232 million (USD 59 million)** in avoided costs over a decade — a conservative *lower bound* that excludes outpatient care, productivity losses, lifetime disability transfers, and the value of infant lives saved.

# Takeaway

- A gradual, phased ban on pre-harvest sugarcane burning — backed by regulation *and* voluntary export-market incentives — delivered large, well-identified health gains
- The pattern across outcomes traces a **dose-response gradient**: null adult mortality → significant infant mortality → significant hospitalizations → significant congenital anomalies, each margin defined by a different vulnerability threshold
- Relevant beyond São Paulo: Northeast Brazil (Pernambuco: >90% manual), India, Thailand, Florida still rely on pre-harvest burning

**Thank you! Questions?**

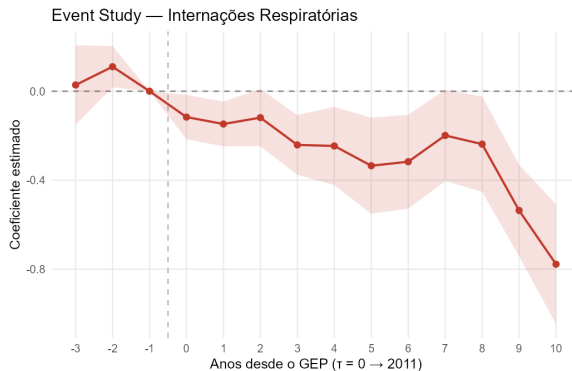
# Backup Slides



# Summary Statistics

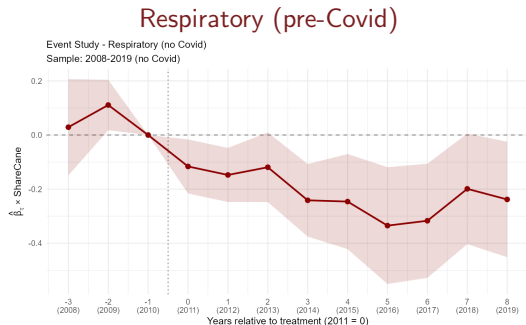
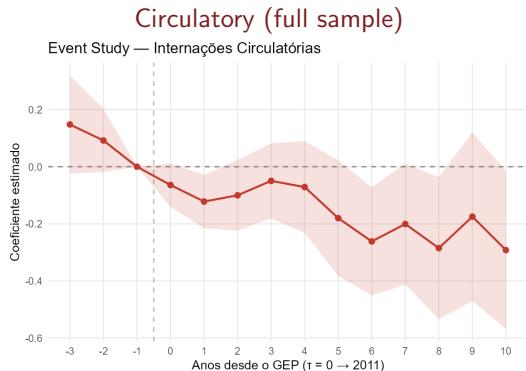
| Variable                                  | <i>N</i> | Mean   | Median | SD     | Max     |
|-------------------------------------------|----------|--------|--------|--------|---------|
| Sugarcane land share (2002–2010)          | 642      | 0.150  | 0.067  | 0.190  | 0.886   |
| Respiratory hosp. per 1,000 pop.          | 8,385    | 7.281  | —      | 4.939  | 45.828  |
| Circulatory hosp. per 1,000 pop.          | 8,385    | 6.470  | —      | 3.085  | 25.760  |
| Any anomaly (per 1,000 births)            | 11,882   | 18.662 | —      | 91.807 | 1,000   |
| Infant resp. mortality (per 1,000 births) | 9,029    | 0.702  | —      | 3.204  | 111.111 |

# Identification: Parallel Trends Hold



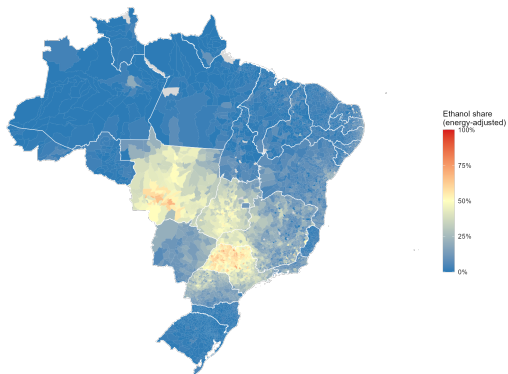
Pre-treatment coefficients ( $\tau = -3, -2$ ) are near zero and insignificant; post-treatment effects grow steadily as the GEP phases in — consistent with a gradual, accelerating rollout, not a one-time break.

# Event Study — Circulatory and Pre-Covid Samples



The same pattern survives dropping the pandemic years, so the result is not a Covid artifact.

## Backup: Ethanol Market Context



Energy-adjusted hydrated ethanol share in the fuel mix, 2020–2024. Hydrated ethanol competes directly with gasoline where transport costs from the cane belt are low.

## Backup: Aggregate Birth Outcomes (Municipality-Year)

|              | $\ln(\text{birthweight})$ | $\ln(\text{LBW})$ | $\ln(\text{VLBW})$ | $\ln(\text{anomalies})$ |
|--------------|---------------------------|-------------------|--------------------|-------------------------|
| Tech. Change | -0.002<br>(0.003)         | 0.027<br>(0.023)  | -0.067<br>(0.056)  | -0.795**<br>(0.311)     |

The aggregate specification confirms the anomaly result; the individual-level LPM shown earlier additionally decomposes it by diagnostic group.

# References I

- André LS Chagas, Carlos R Azzoni, and Alexandre N Almeida. A spatial difference-in-differences analysis of the impact of sugarcane production on respiratory diseases. *Regional Science and Urban Economics*, 59:24–36, 2016.
- C Austin Davis. Why did sugarcane growers suddenly adopt existing technology? Technical report, Working Paper - Yale Department of Economics, 2017.
- Marcos A Rangel and Tom S Vogl. Agricultural fires and health at birth. *Review of Economics and Statistics*, 101(4):616–630, 2019.